

# Lecture 15

## Perturbation Theory for Linear Operators

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### Recommended route

This sheet accompanies the optional, unassessed capstone. Work Exercises 15.1–15.7 for the recommended route. The remaining problems are extensions. Throughout,  $H(\varepsilon) = H_0 + \varepsilon V$  with  $H_0$  and  $V$  self-adjoint and  $\varepsilon \in \mathbb{R}$ , and denominators are written with the level of interest first.

**Exercise 15.1** (computational) Let  $H_0 = \text{diag}(1, 4)$  and  $V = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ . Find the first- and second-order shifts of both levels, and compare the second-order prediction with the exact eigenvalues at  $\varepsilon = 0.1$ .

**Exercise 15.2** (computational) For  $H_0 = \text{diag}(0, 2, 5)$  and

$$V = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 2 \\ 0 & 2 & 0 \end{pmatrix},$$

compute all three first- and second-order shifts, and verify the two checks  $\sum_n E_n^{(1)} = \text{tr } V$  and  $\sum_n E_n^{(2)} = 0$ .

**Exercise 15.3** (computational) For the three-level system of the lecture,  $H_0 = \text{diag}(0, 1, 3)$  with  $V$  having rows  $(2, 1, 1)$ ,  $(1, 1, 1)$ ,  $(1, 1, -3)$ , compute the first-order correction to the *top* eigenvector and evaluate it at  $\varepsilon = 0.1$ .

**Exercise 15.4** (computational) Let  $H_0 = \text{diag}(0, 1, 3)$  and  $V = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ . Compute  $\|V\|$ , and find the largest  $|\varepsilon|$  for which the three disks of the spectral stability theorem are still pairwise disjoint.

**Exercise 15.5** (proof) (*Hellmann–Feynman.*) Let  $|n(\varepsilon)\rangle$  be a normalized eigenvector of  $H(\varepsilon)$  depending differentiably on  $\varepsilon$ , with eigenvalue  $E_n(\varepsilon)$ . Prove that

$$\frac{dE_n}{d\varepsilon} = \langle n(\varepsilon) | V | n(\varepsilon) \rangle,$$

and deduce the first-order formula  $E_n^{(1)} = \langle n | V | n \rangle$ .

**Exercise 15.6** (proof) Assume  $H_0$  has simple eigenvalues. Prove that  $\sum_n E_n^{(2)} = 0$ , and that the second-order shift of the *lowest* level is always  $\leq 0$  while that of the highest is always  $\geq 0$ .

**Exercise 15.7** (computational) Let  $H_0 = \text{diag}(2, 2, 6)$  and

$$V = \begin{pmatrix} 1 & 3 & 0 \\ 3 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Explain why the non-degenerate formulas cannot be used, then find the first-order splitting of the degenerate level and the correct zeroth-order states.

### Further exploration

**Exercise 15.8** (★) For  $\Delta > 0$  consider the two-level family

$$H(\varepsilon) = \begin{pmatrix} \Delta/2 & \varepsilon \\ \varepsilon & -\Delta/2 \end{pmatrix}.$$

(a) Find the exact eigenvalues. (b) Expand the upper one to order  $\varepsilon^2$  and check it against second-order perturbation theory. (c) Regarding  $\varepsilon$  as a complex variable, find the radius of convergence of that expansion, and say what obstructs it.

### Infinite-dimensional outlook

**Exercise 15.9** (computational) Let  $D$  be the diagonal operator  $De_n = \frac{1}{n}e_n$  on  $\ell^2$  and let  $z = -1$ . Compute  $\|(D - z \text{id})^{-1}\|$ , and find the largest  $|\varepsilon|$  for which the resolvent expansion  $(D + \varepsilon V - z \text{id})^{-1} = \sum_{k \geq 0} (-\varepsilon)^k R_0(z) (V R_0(z))^k$  is guaranteed to converge, for a bounded self-adjoint  $V$  with  $\|V\| = 2$ .

**Exercise 15.10** (★) Explain why the Rayleigh–Schrödinger series for the quartic anharmonic oscillator  $H(\varepsilon) = \frac{1}{2}(\hat{p}^2 + \hat{x}^2) + \varepsilon \hat{x}^4$  cannot have a positive radius of convergence, even though its first few terms are numerically excellent. Contrast this with the finite-dimensional situation of Lecture 15.